

Geometrical and statistical analysis of dynamic crack morphology in shrink-swell soils with addition of maize roots or salinity (NaCl)

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ABSTRACT

Cracks generated in shrink-swell soils detrimentally affect hydraulic and mechanical properties of soils, and alter soil-water processes, such as gas diffusion, water flow, solute transport, and root penetration. Root additives and salinity beget physicochemical changes in soils, resulting in variations of crack morphology. Understanding these variations provides effective means of crack utilization or prevention. In this study, we investigated the effects of maize roots and sodium chloride (NaCl) on crack dynamics and morphology in shrink-swell light clays. Drying experiments were conducted in soils containing roots (root densities: 0, 0.5, 1.0 and 1.5 cm/cm³) or NaCl (mass contents: 0.03, 1.0, 1.5 and 2.0 %). Crack evolution patterns were photographed at regular intervals and crack parameters were calculated from the resulting images using morphological algorithms. We investigated the dynamic geometry, topology, and statistical behaviors of cracks. Results indicate that incorporating maize roots effectively restrained crack area and width, but increased crack length, tortuosity, and number of cracks. Crack networks in root-treated soils contained a vast number of dead-ends and displayed pronounced fragmentation and low connectivity. Distribution of crack intersection angle well approximate normality with a mean around 120°. The bonding and friction between roots and particles, the bridging effects, stiffness, and ductility of roots improved the soil tensile strength and altered the intrinsic release criteria of tensile stress. The NaCl treatments increased crack area and width but decreased total crack length and tortuosity. With a slight amount of NaCl, the crack network tended to be highly connected with no dead-ends. Distribution of crack intersection angle in sodic soils exhibited asymmetric bimodality and was well described by a mixture of two Gaussian distributions, with primary and secondary peaks around 105° and 165°, respectively. We proposed a statistical function that describes crack area probability distribution considering water content as a variable, which proved to be a good fit in modeling dynamic statistics of crack patterns. Our results provide compelling evidence of how roots and salinity alter crack evolution, dynamics, and morphology in shrink-swell soils, which will be beneficial in crack inhibition and utilization, and in hydrological use such as water management in saline-sodic soils and prevention of preferential flow. However, cracking behaviors are highly susceptible to sample size effects and basal boundary. We hypothesize cracking behaviors (in roots- or salinity-treated soils) in small scale and in-situ scale are dimensionally similar, and the validity of these conclusions being generalized to in-situ cracked soils should be further investigated.

1. Introduction

Desiccation crack is a phenomenon of physical failure resulting from the sustained development of capillary suction and tensile stress inside the soil matrix upon drying (DeCarlo and Shokri, 2014a; Stewart et al., 2016b). Cracking behavior detrimentally affects both hydraulic and

mechanical properties of soils (Rayhani et al., 2007; Tang et al., 2008). Cracks act as preferential pathways for transport of fluids and contaminants, increasing hydraulic conductivity of soils and causing water and fertilizer to bypass absorption by the surface layer (Greve et al., 2010). Consequently, this results in a decline in irrigation efficiency and possibly induce groundwater contamination. Cracks typically occur in

Abbreviations: CK, control treatment; R1, R2 and R3, root treatments of 0.5, 1.0 and 1.5 cm/cm³; S1, S2, and S3, salinity treatments with mass contents of 1.0, 1.5 and 2.0 %; R2-3, the 3rd replicate in the 2nd root treatment R2, likewise for the other treatments; SKIZ, skeleton by zones of influence; CIF, crack intensity factor.

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shrink-swell soils with high contents of clay minerals, which makes it more complex to predict the fluid transport in shrink-swell soils, since the porous medium turns into a variable-solid-skeleton soil and the cracks are typically randomly distributed (Stewart et al., 2016a; Wang et al., 2018a). Crack morphologies are particularly relevant for understanding soil deformation-shrinkage mechanisms and accurately predicting water-solute transport in shrink-swell soils (Peng et al., 2016), as crack morphology provides geometrical information that acts as boundary conditions for preferential flow (Wang et al., 2018b). The factors that determine the formation of cracks and cracking morphology are of particular interest since crack morphology is hydrologically relevant to preferential flow behavior.

In naturally agricultural soil systems, both intrinsic and external factors dominating soil shrinkage behavior can also dictate crack formation, propagation, and network morphology. The intrinsic factors include soil structure, mechanical composition (mainly clay content), type and amount of clay minerals (Yesiller et al., 2000), organic matter (Preston et al., 1997), salinity (DeCarlo and Shokri, 2014b; Zhang et al., 2016), microbial communities (Preston et al., 2001), and the external factors typically refer to environmental conditions (e.g. temperature and humidity, wetting-drying cycles) (Tang et al., 2008; Wang et al., 2017) and biological intervention (e.g. wormholes, roots, human tillage). Mitchell (2005) reported that desiccation cracking is intrinsically controlled by the type and amount of soil clay minerals owing to their shrink-swell features (Mitchell, 2005). The extent of cracking increases with higher fines content in soil, making the fines content a valid indicator of cracking (Yesiller et al., 2000). After the soil undergoes wetting-drying cycles, irreversible changes occur in the soil fabric, and a weakening in soil strength is observed (Wang et al., 2017; Yesiller et al., 2000). Salt treatment greatly affects soil physical, hydraulic properties, and evaporation behaviors (So and Aylmore, 1993; King et al., 2017). For soils with high sodium adsorption ratio (SAR), sodium is adsorbed and closely attached to soil particles, leading to a breakdown in soil's physical structure (Fipps, 2003). As regards fine-textured soils, clays in soils are most subject to breakdown, thus the shrinkage behavior and tensile strength of these soils are significantly changed. Total salinity, being another salinity hazard to soils, greatly affects total water potential and solid-liquid contact angle, which possibly plays a major role in cracking mechanism. Relatively little research has been done revealing the effects of salinity or sodicity on cracking morphology, and among those existing literature, almost opposite experimental results have been reported. For instance, experimental results conducted by Lima and Grismer (1992) reported that as salinity (NaCl) increased, soil crack area, width, and crack volume tended to increase, which can be attributed to a lack of macropores in salt-affected soils. While DeCarlo and Shokri (2014b) and Zhang et al. (2016) reported partially opposite conclusions, stating that with the increase in NaCl, crack intensity factor (CIF), intersection number and total crack length decrease. Apart from the basic geometrical parameters, e.g. crack area, width and length, there has been little quantitative analysis of crack morphology (e.g. network topology) as affected by salinity which associates with stress-energy release and cracking mechanism.

As mentioned above, shrink-swell clayey soils are prone to cracking upon drying, and cracks tend to increase the risk of groundwater contamination and preferential flow. Therefore, the amendment of soil swelling-shrinkage and cracking dynamics is of great interest to agricultural sustainability and production (Zong et al., 2014). Additives used to amend soils involve natural plant roots, available fibers such as straws and vegetable fibers, for the sake of enhancing soil strength and suppressing desiccation cracks (Chaduvula et al., 2017; Xing and Ma, 2019). Fiber amendment aids in inhibiting or retarding tensile crack formation, and in reducing soil swelling potential and thermal conductivity (Chaduvula et al., 2017). It is also documented that straw residues favor conserving nutrients, increasing carbon organic matter, regenerating soil structure, and modifying soil physical and hydraulic properties, which benefits sustaining production in rice-wheat cropping

systems (Singh et al., 2005). Addition of biochar (produced from pyrolysis of wood at 500 °C into soils reduces the degree of soil cracking and keeps the soil intact, as reported by Zhang et al. (2020). Bordoloi et al. (2018) also reported that, with the inclusion of water hyacinth biochar, the stabilized crack intensity factor decreases from 7 % to 2.8 %, which could be attributed to the less cohesion of biochar particles than soil fines (Bordoloi et al., 2018). There is no shortage of literature on how straws, roots, and fibers affect soil shrinkage, whereas little has been reported regarding the dynamic cracking morphology altered by root additives, which associates closely with cracking mechanism and is an essential prerequisite for predicting preferential flow in cracked soils with roots.

The main objective of this study is to analyze the geometrical, topological, and statistical characteristics of crack morphology in light clay soils upon drying, as affected by maize roots and NaCl addition. With the aid of digital image processing and morphological algorithms, this study aims to (1) investigate how roots treatments affect the geometrical and statistical features of crack, and modify the crack morphology since roots noticeably alter the soil tensile stress field; and (2) uncover how sodicity (NaCl) affects cracking mechanism through analysis of crack patterns (i.e. crack geometry and morphology), given the known effects of sodium on the aggregate structure and soil swelling behavior.

2. Materials and methods

2.1. Soils

Surface samples (0–20 cm) were taken from the experimental fields in the Key Laboratory of Efficient Irrigation-Drainage and Agricultural Soil-Water Environment in Southern China, Ministry of Education. The soil is a light clay according to ISSS soil texture classification and is categorized as Hapludalf, Alfisols. The properties of the soil are presented in Table 1. Sample preparation and protocol are similar to Wang et al. (2017). The soils were air-dried, sieved through a 2-mm-aperture screen mesh, and stored in a well-ventilated lab (25 ± 2 °C, 50 ± 1 % relative humidity) before use.

2.2. Treatments

Maize roots from milk stage were used as root treatment in this study. Maize with the entire root system was taken from the experimental field (wheat-maize rotation system) and the residual soil around the roots was removed. The maize roots were cut into 5-cm-length segments (roots ranged in diameters from 0.3 to 0.7 cm) and incorporated into the sieved soils. The root length density (unit: cm/cm³) here refers to the ratio of root length to the soil volume. The root treatments consisted of three levels of root densities: 0.5, 1.0, and 1.5 cm/cm³, which correspond to maize root contents in jointing stage, milk stage, and physiological maturity stage, respectively. A control treatment was set without any root treatment (CK). The experiments were performed in square Perspex

Table 1
Physical properties of the tested soil.

Soil properties	Value
Type	Light clay (ISSS soil texture classification)
Classification	Hapludalf, Alfisols (yellow-brown earth)
Sand content (>0.02 mm), %	19.68
Silt content (0.002 mm–0.02 mm), %	36.41
Clay content (<0.002 mm), %	43.91
Dry bulk density, g cm ⁻³	1.41
Field capacity, %	31.4
Saturated hydraulic conductivity, cm h ⁻¹	3.826
Specific gravity	2.65

plates ($22 \times 22 \text{ cm}^2$). Once thoroughly mixed with roots, the soil sample was packed into the Perspex plates to a bulk density of 1.41 g/cm^3 for later desiccation experiments.

The other series of treatments was to pretreat the sieved soils using sodium chloride solutions (NaCl: analytical grade and purity higher than 99.5 %) with different concentrations. Saline soils were made by sprinkling a set of 640 g sieved soils with 400 mL distilled water or NaCl solutions and mechanically stirring once a week to ensure a homogenous mixture. The added salts were 0, 6.4, 9.6, and 12.8 g, and through conversion, the molar concentrations of the solution were 0, 0.274, 0.410, and 0.547 mol L^{-1} , respectively. All samples were kept in a well-ventilated environment for 30 days and were then packed into Perspex plates to a bulk density of 1.41 g/cm^3 . The samples treated with distilled water were set as control treatments (CK) and its salinity mass content approximated 0 (around 0.03 %). Through calculation, the salinity mass contents (mass ratio of salinity to the original sieved soil) of the treatments are 0.03 %, 1.0 %, 1.5 %, and 2.0 %. Three replicates of each root and salinity treatment were prepared. The details of roots and salinity treatments are summarized in Table 2.

2.3. Generation of cracking patterns

After packed into the Perspex plates, each soil sample was oversaturated with 400 mL distilled water and then naturally dried in a temperature-humidity controlled room ($35 \pm 2^\circ\text{C}$, relative humidity $60 \pm 5\%$). During drying, the sample was periodically weighed at 4-hr intervals using a digital balance (accuracy: 0.1 g) to calculate evaporation and gravimetric water content. Meanwhile, crack morphology was recorded on a digital camera (resolution: 4608×3408) mounted above the Perspex plate. The samples were considered to be fully desiccated until the difference between two consecutive weights was less than 0.2 %. When the soil surface was photographed, we illuminated it by an LED surrounding the Perspex plate to optimize the quality of images and to facilitate image processing and morphological analysis. The cracking process typically continued for 5 days. After that, we processed the crack images and analyzed the crack morphology with the aid of image processing techniques.

2.4. Image processing and morphological algorithms

Image processing was performed in batch mode using MATLAB R2016a according to the methodology introduced in Wang et al. (2018b). The original images were first cropped to obtain the region of interest (core $20 \times 20 \text{ cm}^2$, object crack pattern) and to minimize edge effects, and then resized to a resolution of 2048×2048 pixels. Subsequently, the cropped color images were converted to gray format (gray levels 0–255). The contrast between cracks and shrinking clods (backgrounds) in grayscale was marked so that a standard thresholding method was sufficiently enough to yield a binary image that comprised only cracks and backgrounds. In our study, we employed the OTSU thresholding method which identifies an optimal threshold (T) by

minimizing the ratio of between-class variance to the total variance (Parker, 2010; Wang et al., 2018b). Thus, pixels with gray levels below T were classified as cracks and those with gray levels above T were shrunk clods. After that, spots in binary images were removed, and gaps and broken features were reconnected, using ‘opening’ and ‘closing’ operations, respectively.

Crack binary image was smoothed, and then skeletonized using a parallel thinning algorithm (Zhang-Suen algorithm) that iteratively removes the outermost pixel layers that meet the criteria of discrimination (Wang et al., 2018b). The crack length was approximated based on the resulting crack skeleton. The skeleton was one-pixel wide, and the skeletal pixels are horizontally, vertically, or diagonally connected. Typically, skeletal spurs were frequently caused by crack edge irregularities. We used edge smoothing prior to skeletonization to reduce the number of spurious pixels. If spurs remain, they were eliminated using MATLAB internal function ‘bwmorph (spur)’. We scanned the crack skeleton and each skeletal pixel was identified as an endpoint, trunk point, or branchpoint using the connectivity number of each skeletal pixel in a 3×3 pixel region (Wang et al., 2018b).

Crack networks can be regarded as a pattern of intersecting cracks at branch points. To analyze individual crack geometry, cracks were separated at branch points, based on the skeleton by zones of influence (SKIZ) and watershed algorithm (Wang et al., 2018b). The method for crack separation is as follows:

- (1) Prepare crack binary image, skeleton and branch points images;
- (2) Dilate branch points with a square structuring element;
- (3) Subtract the dilated branch points from the skeleton;
- (4) Perform Euclidean distance transform on the skeleton without branch points, thus producing the elevation map in which each pixel was assigned the Euclidean distance from the nearest skeletal pixel;
- (5) Perform watershed segmentation on the elevation map, thereby producing the watershed lines and catchment basins with regional minimums;
- (6) Identify 0-valued pixels (watershed lines) as SKIZ;
- (7) Subtract SKIZ from the binary image.

The whole process is illustrated in Fig. 1. Crack images were processed through batch mode in a MATLAB script in which programmed functions and intrinsic functions from the image processing toolbox were called. The methodology yielded crack binary images, skeleton, branch points and endpoints, and separated cracks.

2.5. Crack parameters

We used morphological algorithms to measure crack patterns at pixel scale, based on the resulting crack binary image, skeleton, nodes, and separated cracks. The parameters used to characterize crack morphology include crack area ratio, length density, mean width, tortuosity of individual crack, crack intersection angle; and the topology of a crack network is described by crack network connectivity, box-counting fractal, and crack intersection angle.

Crack area ratio, R_{cr} , refers to the time-variable ratio of the superficial crack area to the region of interest and was calculated as the count of crack pixels divided by the count of all pixels in the binary image. This index, also referred to as Crack Intensity Factor (CIF), was first proposed by Miller et al. (1998) and used as descriptor of extent of surface cracking with drying time.

Crack length density, l_{cr} , is defined as the crack length per unit area. The calculation of crack length made use of the criterion of ‘8-connectivity’ in digital morphology. As mentioned above, skeletal pixels in 8-connected regions are either orthogonally (vertically-horizontally) or diagonally connected. Pixel length of an orthogonal joint is 1 and that of a diagonal joint is $\sqrt{2}$. The crack length was approximated by summing

Table 2
Roots and salinity treatments.

Specimen	Treatment	Density (cm cm^{-3})	Length of a root segment (cm)
CK	Maize roots	0	5
R1	Maize roots	0.5	5
R2	Maize roots	1.0	5
R3	Maize roots	1.5	5
Specimen	Treatment	Mass content (g g^{-1} 100 %)	Molar concentrations of solution (mol L^{-1})
CK	Total salts	0.03 %	–
S1	NaCl	1.0 %	0.274
S2	NaCl	1.5 %	0.410
S3	NaCl	2.0 %	0.547

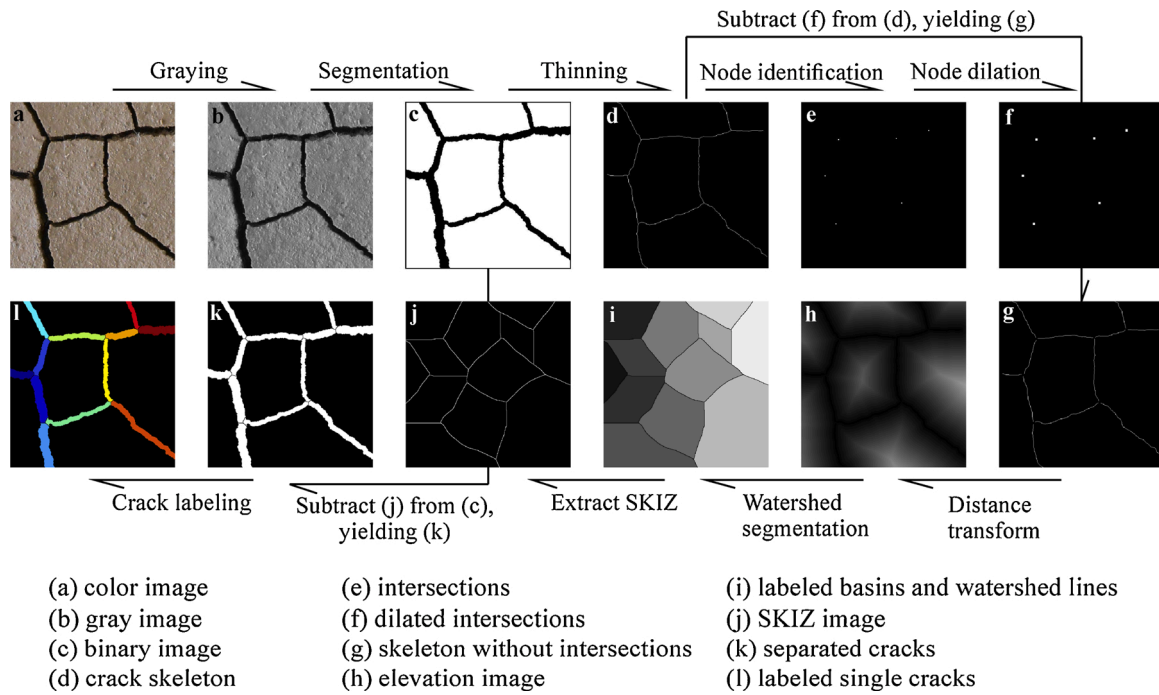


Fig. 1. Flowchart of crack image segmentation (a–c), crack thinning (c–d), node identification (d–e) and crack separation (c–l).

up all lengths between every two joint skeletal pixels:

$$L_{cr} = \lambda_L \left[n_{orth} + \sqrt{2} n_{diag} + \frac{(1 + \sqrt{2})}{2} n_{iter} \right] \quad (1)$$

where n_{orth} and n_{diag} are the counts of orthogonal and diagonal joints, respectively. During skeletonization, iterative removal of spurs on the skeleton led to an underestimation of crack length at endpoint vicinity. An approximate correction was applied to the crack length by adding an extra term, namely the number of iterations (n_{iter}) for spur removal multiplied by the mean length between skeletal neighbors. λ_L denotes the unit conversion from pixel to length.

Mean crack width, w_{cr} , was calculated as the crack area divided by the crack length.

Tortuosity of an individual crack, τ_{cr} , characterizes the curving feature of a single crack and was calculated as the ratio of the actual crack length to the straight distance between the start and endpoints of the single crack (Luo et al., 2010). A value of 1 describes a straight crack, and as the value increases the crack becomes increasingly tortuous. Each crack is extracted from the image of separated cracks.

Crack intersection angle, β_{2cr} , refers to the angle between two intersecting cracks. This measure is closely associated with crack propagation and stress release criterion (Vogel et al., 2005) and is measured based on the images of the crack skeleton and branch points. For each branchpoint, we placed an 11×11 pixels region superimposed on the crack skeleton and centered at the branchpoint. Typically, three skeletal pixels overlapped with the outermost layer of the pixel region. Each edge skeletal pixel and the central branchpoint formed a line, and the angle between every two lines was regarded as a crack intersection angle.

Crack network connectivity, c_{net} , is a measure of the state of crack network being connected. This index dominates the connected pathways along which gravity-driven flow can be transported through soils (Katuwal et al., 2015). In this study, the connectivity of a superficial crack network was quantified based on the counts of branch points and endpoints, which was calculated using:

$$c_{net} = \frac{n_{brp}}{n_{endp} + n_{brp}} \quad (2)$$

where n_{brp} and n_{endp} are the counts of branch points and endpoints, respectively.

Fractal dimension, D , provides a quantitative measure of self-similarity, space-filling ability, complexity, and/or tortuosity of fractal structures (Perret et al., 2003). Crack network propagated randomly and hierarchically, which is suggestive of fractal behaviors. In this study, we use box-counting fractal (D_B) to characterize the scaling features of cracks, which is widely adopted due to its simplicity of mathematical calculation or empirical estimation (Ai et al., 2014; Dathe and Thullner, 2005; DeCarlo and Shokri, 2014a; Liu et al., 2005). D_B was measured by superimposing square boxes of side length δ onto the binary crack image and recording the count of boxes $N(\delta)$ that overlapped any part of cracks. A sequence of boxes of descending sizes was selected, and the box-counting fractal was determined as the slope of $\log N(\delta) - \log \delta$ plot using:

$$\log N(\delta) = -D_B \log \delta + c \quad (3)$$

where c is a constant. Selections of box size depend on the resolution of binary crack image. In our study, the binary image size is 2048×2048 pixels and we defined the serial box sizes as $\delta = 2^{i-1}$, where $i = 1, 2, \dots, 11, 12$.

2.6. Statistical analysis

We specially analyzed the desiccated crack patterns of all root and salinity treatments. For each crack pattern, measures including crack length, crack area, tortuosity of individual cracks, crack intersection angle were calculated, and the frequency distributions of these measures were generated. We compared the distributions with classical probability density functions. Three models, namely, normal distribution, lognormal distribution, and bimodal distribution were found as good model candidates to fit these distributions. Mathematically, the bimodal distribution is an addition of two overlapping Gauss distributions. The probability density functions of the three distributions are given in Table 3. The function parameters were approximated using Maximum Likelihood Estimation, and the goodness of fit was evaluated using adjusted R^2 (coefficient of determination).

Table 3
Probability density functions of three candidate distributions.

Distribution	Parameters	Probability density function
Normal	μ, σ	$f(x) = \frac{A}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Lognormal	μ, σ	$f(x) = \frac{A}{\sqrt{2\pi}\sigma x} e^{-\frac{(\ln x/\mu)^2}{2\sigma^2}}$
Bimodal	μ_1, σ_1	$f(x) = A_1 f_1(x) + A_2 f_2(x) = \frac{A_1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} +$
	μ_2, σ_2	$\frac{A_2}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x-\mu_2)^2}{2\sigma_2^2}}$

The validity of the three hypothetical distributions was assessed using the Chi-square test. The null hypothesis H_0 that there was no significant difference between the observed and expected distributions was rejected if the chi-statistic was greater than the critical value determined by a significance level of 0.05. Conversely, a smaller Chi-statistic than the critical value indicates an acceptance of the null hypothesis H_0 , suggesting a good fitness between the observed and expected distributions (Li and Zhang, 2010). The Chi-statistic is written as:

$$\chi^2 = \sum_{i=1}^k \frac{(F_i^o - F_i^e)^2}{F_i^e} \quad (4)$$

where χ^2 is a random variable that follows $\chi^2(k-1)$ distribution with $k-1$ degrees of freedom, based on the premise that the sample size (the number of observations) is large enough (>50). k is the number of bins (frequency classes), F_i^o and F_i^e are the observed and expected frequency at class i , respectively. If $\chi^2 < \chi_{\alpha}^2(k-1)$, the hypothesis is accepted, in which α is the significance level. Typically, a value of 0.05 is assigned to α . The critical value $\chi_{\alpha}^2(k-1)$ can be determined by consulting the Chi-square distribution table. All the statistical analyses were done in a programed MATLAB script and OriginLab R2017.

3. Results

Crack patterns of desiccated soil (in binary format) under root and salinity treatments are illustrated in Fig. 2. Qualitative observations of

these crack patterns indicate a strong dependence of crack morphology on root densities and salinity contents. With denser roots additives, soil samples display narrower cracks, smaller shrinking clods, and less connectivity of crack networks. The samples with higher root contents contain a large number of dead-end cracks with sharp spikes. This indicates that the evolution of crack network connectivity and crack width was inhibited by root incorporation. The salinity samples exhibit quite different characteristics of crack morphology in contrast to root treatments. With the increase in salinity, cracks become considerably wider, and the crack network propagates more connected and contains scarcely any dead-end branches. The crack networks also exhibit pronounced hierarchy in comparison with the CK and root treatment samples.

3.1. Geometrical characterization of crack patterns

The evolutions of crack area ratio (R_{cr}), length density (l_{cr}) and mean width (w_{cr}) versus gravimetric water content for all root and salinity treatments are plotted in Fig. 3, with the control treatment (CK) marked in hollow symbols being the reference. Fig. 4 shows the final R_{cr} , l_{cr} , and w_{cr} in fully desiccated crack patterns. The water contents mentioned are all gravimetric. Towards the left the samples were dehydrated and the water contents decreased. For all insets, cracks initiated and progressed when the water content was approximately 37.5 %. The crack ratio shows a steady increase with decreasing water content and ultimately stabilizes at 5 % water content, while the crack length density displays a different increasing trend. A steep increase followed by a slight increase in crack skeleton length is observed with decreasing water content. The threshold of the slopes is near 27 % water content, suggesting that the crack skeleton approached stabilization much earlier than the crack area or crack growth.

For root treatments, the crack area ratios are not significantly different among root densities (Fig. 3a–c). As regards the crack length and width (Fig. 3g–i, and m–o), an upward trend in crack length density is observed with increasing root density, while the average crack width decreases with higher root density. These phenomena agree well with the experimental results conducted by (Miller and Rifai, 2004; Xue et al., 2014), in which polypropylene fibers and straw fibers were used as additives. Overall, cracks generated by soils treated with maize roots, polypropylene fibers, and rice straws have significantly higher lengths of crack skeleton and lower width, while no evident trends are observed in crack area ratio according to the data presented in Fig. 4a.

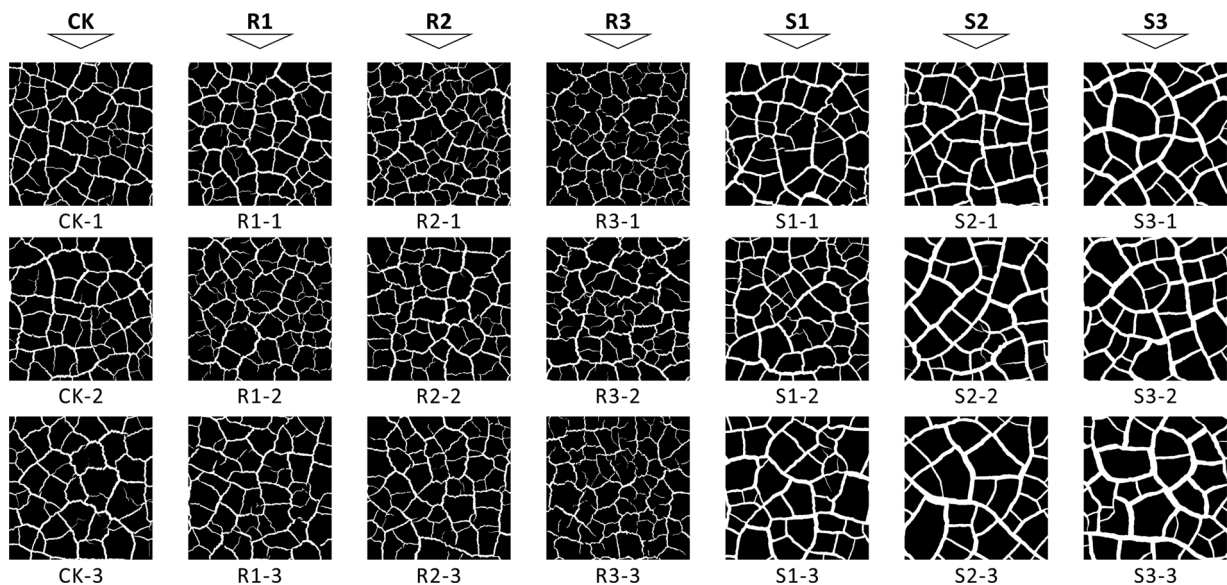


Fig. 2. Binary images of crack patterns in desiccated soil for all root and salinity treatments. ‘R’ and ‘S’ denote root and salinity, respectively. Each column presents three replicates for one treatment.

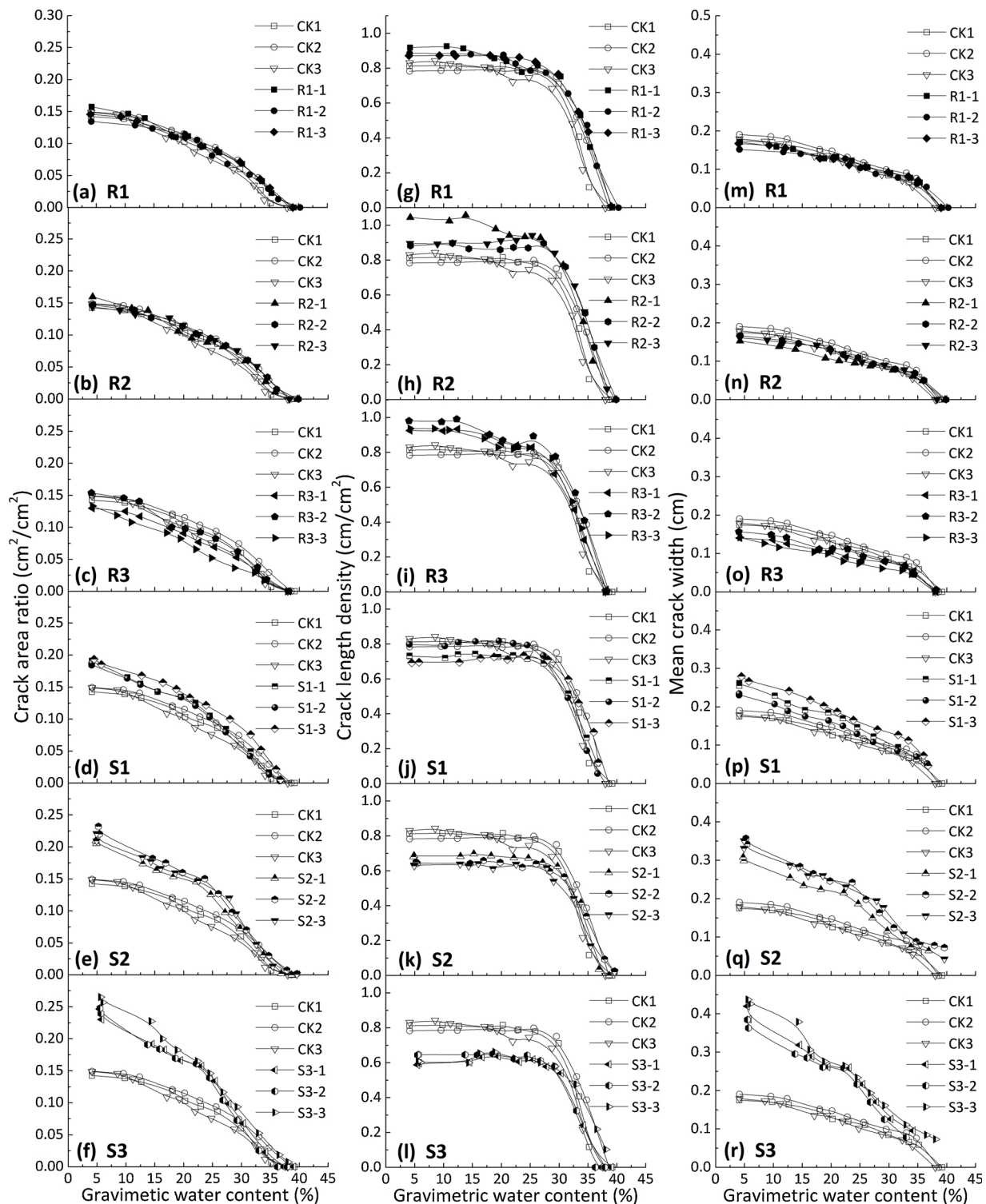


Fig. 3. Variation of crack geometry (incl. crack area ratio, length density and mean width) with gravimetric water content during desiccation for all root and salinity treatments.

Salinity effects on soil cracking geometry are particularly noticeable, as shown in the nine lower insets of Fig. 3. The relationship between crack geometry and salinity content shows that increasing salinity content from 0.03 % to 2.0 % significantly increases the crack area ratio and crack average width, from 14.93 % to 24.89 % and 0.198 to 0.396 cm (data from stabilized crack patterns), respectively. While the crack length density decreases from 0.791 to 0.612 cm/cm². Overall, a linearly upward trend in crack area ratio and crack width is observed with

increasing salinity and the trend of crack length density is reverse.

3.2. Topological characterization of crack networks

The connectivity and tortuosity of soil pores exert an important control on soil hydraulic properties and preferential flow regimes (Vogel and Roth, 2001). The connectivity of the crack network determines the gravity-driven flow pathways and thereby controls the dispersal of

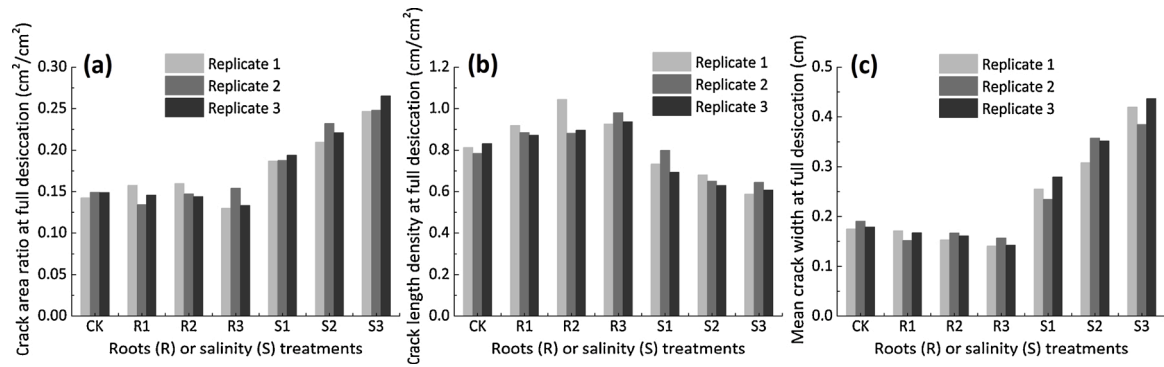


Fig. 4. Area ratio, length density and mean width of cracks in stabilized crack patterns for roots and salinity treatments.

substances (e.g. water, solute, and contaminants) in soils (Jarvis et al., 2017). Fig. 5(a)–(f) shows the growing connectivity of the crack network in soils treated with roots and salinity. Compared with the CK treatment,

a marked decrease in the connectivity of the crack network is observed with increasing root density. The samples with higher roots contents contain a larger number of dead-end cracks and spikes (Fig. 2) and the

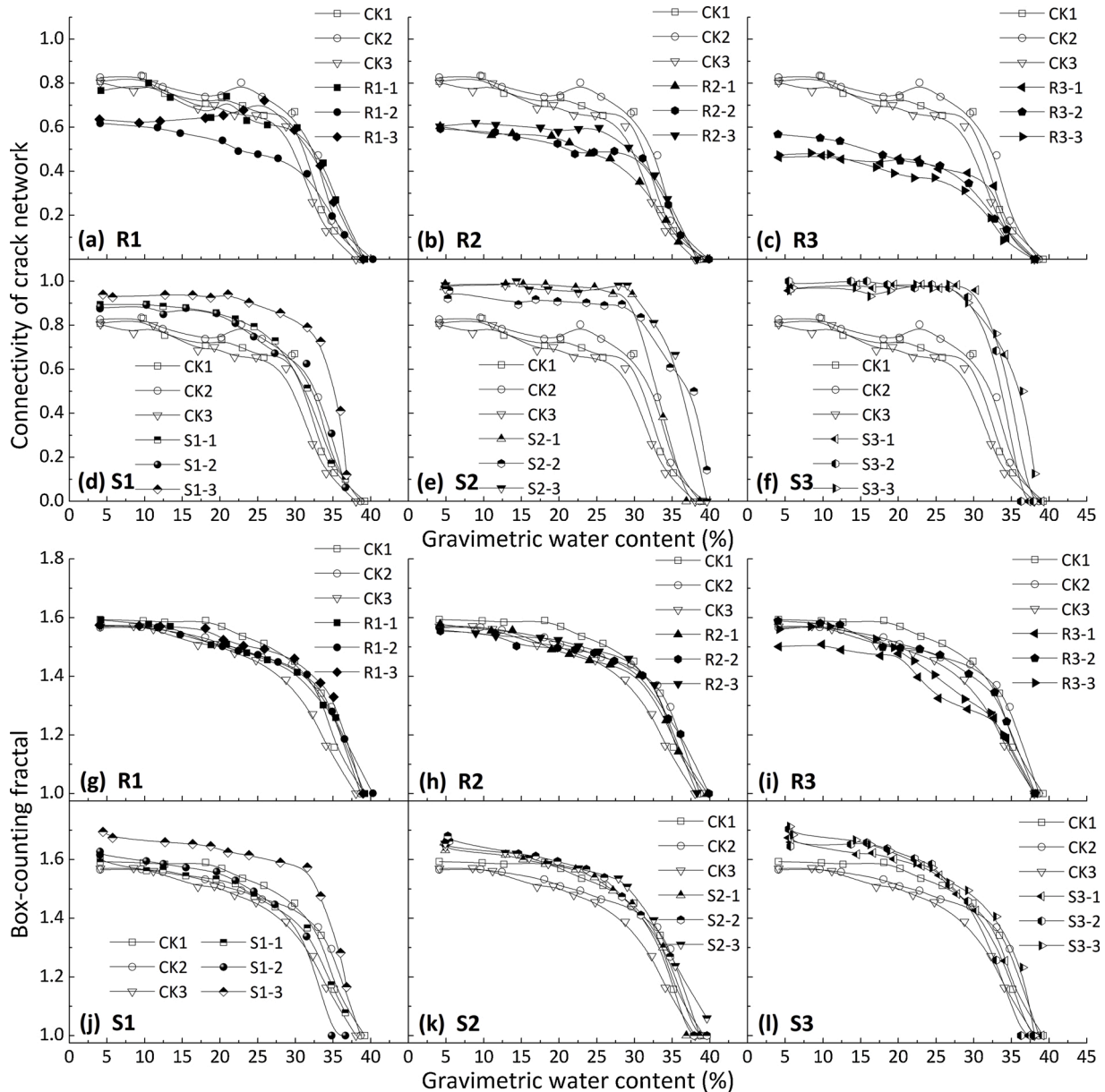


Fig. 5. Variation of crack network topology (connectivity and box-counting fractal) with gravimetric water content for all root and salinity treatments.

cracks in soils treated with roots exhibit a more fragmented morphology.

As shown in Fig. 5(d)–(f), salinity treatment results in higher connectivity of crack networks. Soils that contain merely a slight amount of salinity generated highly connected crack networks and no obvious differences in crack network connectivity are observed among treatments S2 and S3 with salinity mass ratios of 1.5 % and 2.0 %, respectively. The trends of connectivity between CK and salinity-treated samples during the drying process are also distinctly different. For salinity-treated samples, connectivity of crack network first rapidly increases and then converges to approximately 1.0 at a water content of 27

%, which is consistent with the characteristic of crack skeleton growth. The connectivity of CK and all root-treated samples first steeply and then gradually increases, and they follow the same growth behavior. Note that, this measure simply describes the connectivity degree of superficial crack networks. The results, as we assume, could be probably generalized to the superficial crack morphology in a 3-D crack network (e.g. in situ). The high connectivity would predominantly facilitate water flow connectivity and strengthen the preferential flow triggered by cracks. See details in the Discussion section.

Box-counting fractal analysis has been applied to crack networks for

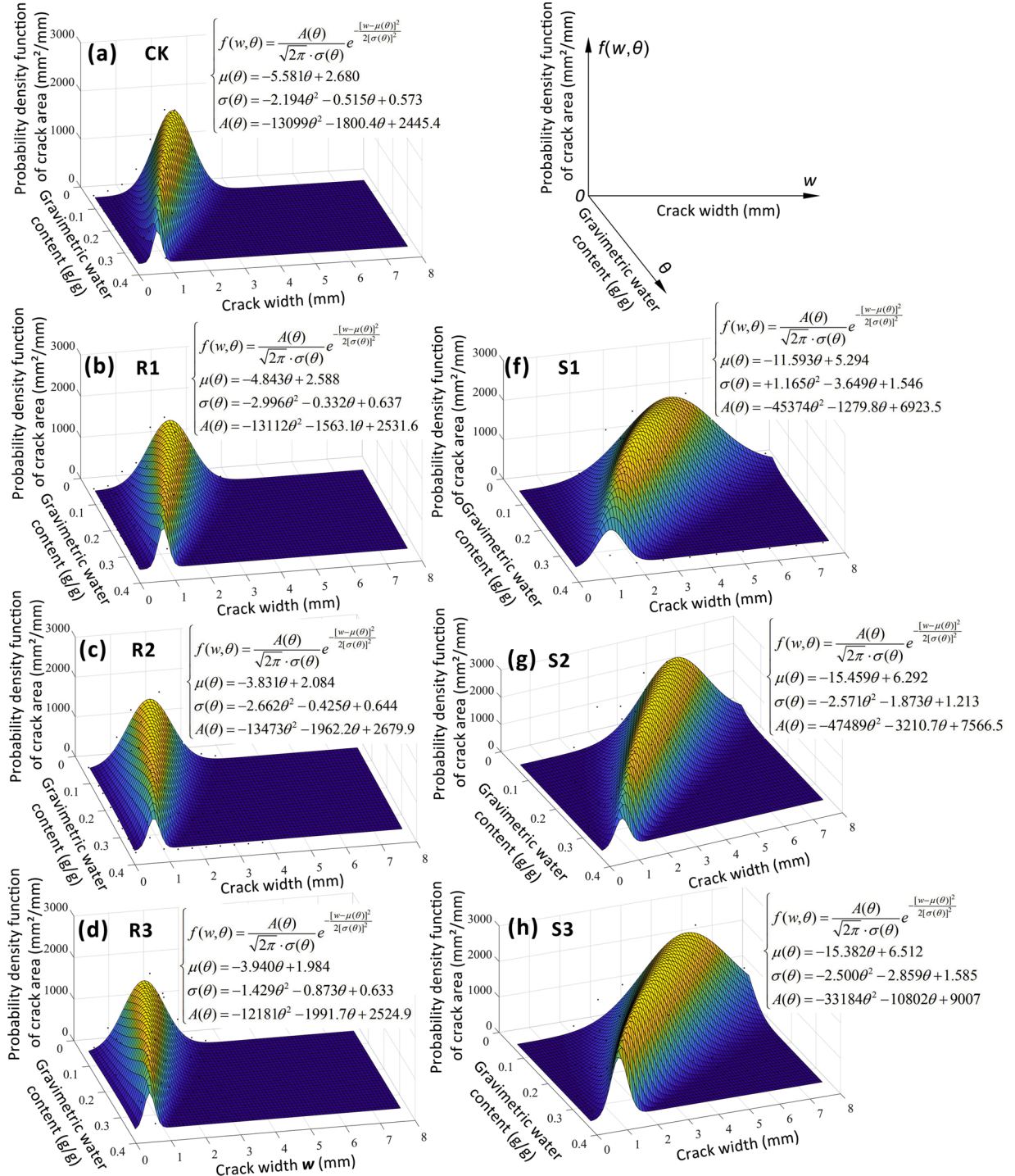


Fig. 6. Probability density function of crack area derived as a function of crack width and gravimetric water content for all root and salinity treatments. The dark dots around 3Ds-surfaces are the observed data. Corresponding expressions are attached.

root and salinity treatments to reveal the growing complexity and self-similarity of the networks, as depicted in Fig. 5(g)–(l). Approximation of the box-counting fractal was done from the regression fitting (see Eq. 3), and high coefficients of determination (all $R^2 > 0.99$) achieved for the fits demonstrate fractal features of crack patterns. Results show that, towards drying, the fractals of crack patterns gradually increase from 1.0 to approximately 1.6. All the fractals of stabilized crack patterns vary between 1.55 and 1.7. Although there are significant differences in crack morphology for all root treatments, the fractals of crack patterns generated by soils with varying root contents follow the same upward trend, and the curves of fractals practically coincide. Likewise, there are also no significant differences in the fractals of crack patterns generated by soils treated with salinity, revealing an analogous degree of self-similarity of crack patterns for all root and salinity.

3.3. Probability distributions of crack area upon drying

For each crack image, we divided crack widths into classes and computed the cumulative crack area within each crack width range. The probability distributions of crack area with respect to crack widths were then analyzed. We found that the crack area approximates a normal distribution across crack width classes (see the equation of normal distribution in Table 3). For all crack images under different water contents, normal distribution is applicable to all crack area distributions. Considering soil water content as a variable, we proposed a probability density function that describes the θ -variable probability distribution of the crack area as a function of crack width w and water content θ . The statistical function was a curved face composed of a parallel displacement of varying Gaussian distributions, and is expressed as follows:

$$f(w, \theta) = \frac{A(\theta)}{\sqrt{2\pi} \cdot \sigma(\theta)} e^{-\frac{(w-\mu(\theta))^2}{2[\sigma(\theta)]^2}} \quad (5)$$

where θ is soil gravimetric water content; w is the crack width; $\mu(\theta)$ and $\sigma(\theta)$ are the mean crack width and standard deviation of a normal distribution at a certain water content θ , respectively; $A(\theta)$ is the area parameter that reflects the total crack areas under different water contents. Parameters $\mu(\theta)$, $\sigma(\theta)$ and $A(\theta)$ are all one-variable functions of θ , solely dependent on moisture content. For a given θ , parameters μ , σ and A could be estimated by fitting the data to a normal distribution, and determinations of all μ , σ and A values at various θ yield correlation functions between the three parameters and water content (θ), respectively. The trends of fitted curves show that parameter- μ is linearly correlated with θ , and parameter- σ and A are quadratically correlated with θ .

Fig. 6 shows the 3D surface of the probability density function (Eq. 5) for typical treatments (CK-3, R1-1, R2-1, R3-2, S1-3, S2-1, S3-3). In each inset, the 3D surface is a parallel of sequential normal curves with various means and variances. The σ -parameter describes the spread of crack widths, with large values indicating that there is a wide range in crack widths. For all root and salinity treatments, both μ and σ increase with decreasing θ , indicating that the mean crack width and the range of crack widths increased during the dehydration process. Also, the estimated μ increases linearly with decreasing θ , which is in good agreement with the experimental data in Fig. 4. The proposed statistical function exhibits pronounced distinctions between roots and salinity treatments. Analysis of the normal curves where θ reaches minimums indicates that the mean crack width (μ) decreases with increasing root contents and increases with higher addition of salinity. The σ in root treatments remains consistent relative to CK, while σ in S1–S3 treatments greatly increases both with increasing salinity contents and with decreasing θ . The goodness of fit was judged by the coefficients of determination (R^2). Results show that the proposed probability density function well describes the crack area distribution across crack width classes under all water contents, with R^2 0.954, 0.921, 0.948, 0.942, 0.894, 0.923, and 0.914 for CK, R1, R2, R3, S1, S2, and S3, respectively.

3.4. Probability distributions of crack measures for stabilized crack patterns

We analyzed the probability distributions of single crack area, length, and tortuosity and crack intersection angles for stabilized crack patterns. Three hypothetical distributions, i.e. normal, log-normal, and bimodal distribution, were selected as three optimal candidates to fit the data. The fitted parameters, e.g. mean values μ , were given in Fig. 8. As shown in Fig. 7(a-1–a-7, b-1–b-7), denser roots added to the soil resulted in more crack segments and cracks with shorter lengths and smaller areas. The distribution of single crack area and length for root treatments is positively skewed, which is broadly consistent with the results conducted by Li and Zhang (2010). In salinity treatments, the areas and lengths of individual cracks are more widely and evenly distributed compared with CK and root treatments. With higher contents of salinity, peak frequency decreases, and values of single-crack area and length corresponding to these peaks increases. The results demonstrate that increasing roots contents resulted in more cracks segments, a broadly decreasing trend in single-crack length and area, while higher addition of salinity led to a pronounced increase in single-crack length and width, a wider range of crack length and area distribution with lower frequency.

The distribution of single-crack tortuosity for root and salinity treatments can be characterized by normal distributions (Fig. 7c-1–c-7) and the fitted means are shown in Fig. 8a. The results suggest that, with root incorporated, the tortuosity of single cracks is significantly higher than CK treatment, with means of 1.103, 1.100, and 1.110 for R1, R2, and R3. The crack tortuosity distribution for root treatments shows a wide range from 1.0 to 1.8, compared with CK and salinity treatments. Results also show that salinity treatment markedly decreases crack tortuosity, with the standard deviation of crack tortuosity significantly lower than root treatments. Note that the tortuosity of a single-crack depends highly on the crack thinning algorithm, and the results focus on the variations in the crack tortuous behaviors with root and salinity added.

The distributions and Gauss fitting curves of crack intersection angles (β_{2cr}) for root and salinity treated samples are shown in Fig. 7(d-1)–(d-7). The distributions of β_{2cr} for CK and root samples well approximate normality, and determinations of means via Gauss fit yield average values ($n = 3$ replicates) of 119.6°, 120.3°, 119.1° and 121.2° for CK, R1, R2, and R3 treatment (Fig. 8b). The distributions of β_{2cr} are not significantly different among CK and root treatments, suggesting that for the tested light clay cracks propagate and intersects non-orthogonally with others predominantly at 120°, which can be interpreted by the criteria of most efficient dissipation of stress. The addition of roots shows no obvious effects on the intersection angles between cracks. Interestingly, the β_{2cr} distribution for all salinity samples shows clearly an asymmetric bimodality. A bimodal function, reproduced by superposing two Gauss probability density functions (Table 3), fits the data well, with yielded asymmetric peaks, i.e. primary and secondary peaks around 105° and 165°, respectively. This agrees well with the observations of the salinity-effected cracking behavior that cracks frequently intersect in ‘T’ and ‘+’ shapes. As reported by Vogel et al. (2005), crack intersection behavior is fundamentally governed by the mechanisms of crack formation and efficient dissipation of energy. Detailed discussion is provided in the subsequent section.

Besides the adjusted coefficient of determination (Adj. R^2), another method we use to validate the candidate statistical distributions is the Chi-square test (χ^2). Normal and bimodal (a mixture of two normal distributions) distributions were tested, as shown in Table 4. Results show that normal and bimodal distributions were appropriate for 92.9 % of the datasets, signifying that the candidate models were broadly valid. Exceptions were the distributions of crack tortuosity for CK-2, R3-3, and S2-2, which could be attributed to the limited amounts of statistical sample and the accidental errors partially caused by digital image processing and morphological methods.

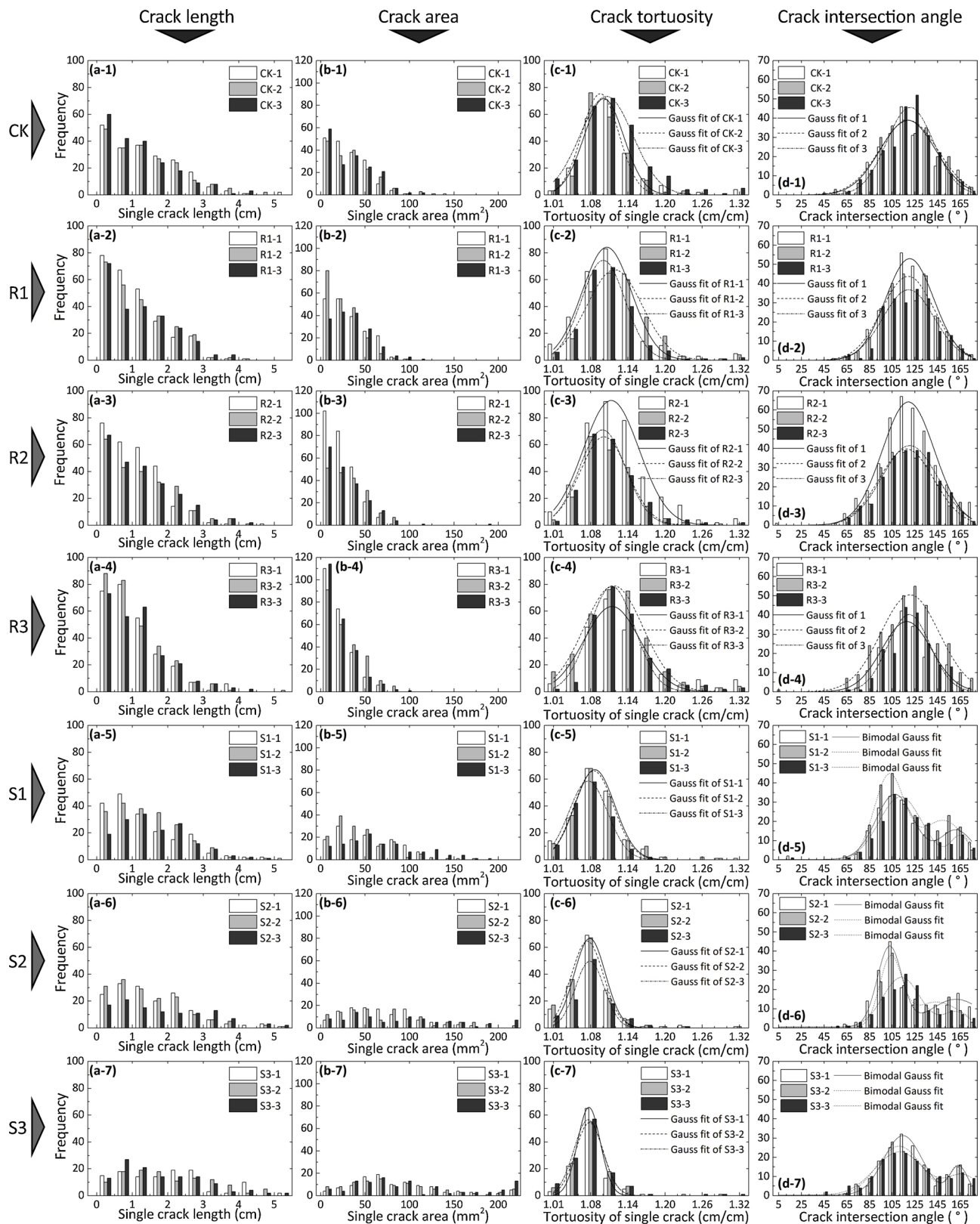


Fig. 7. Distribution of single crack length (a-1-a-7), area (b-1-b-7), tortuosity (c-1-c-7), crack intersection angle (d-1-d-7) and corresponding fitted probability density function for the control treatment (CK), root treatments (R1-R3) and salinity treatments (S1-S3).

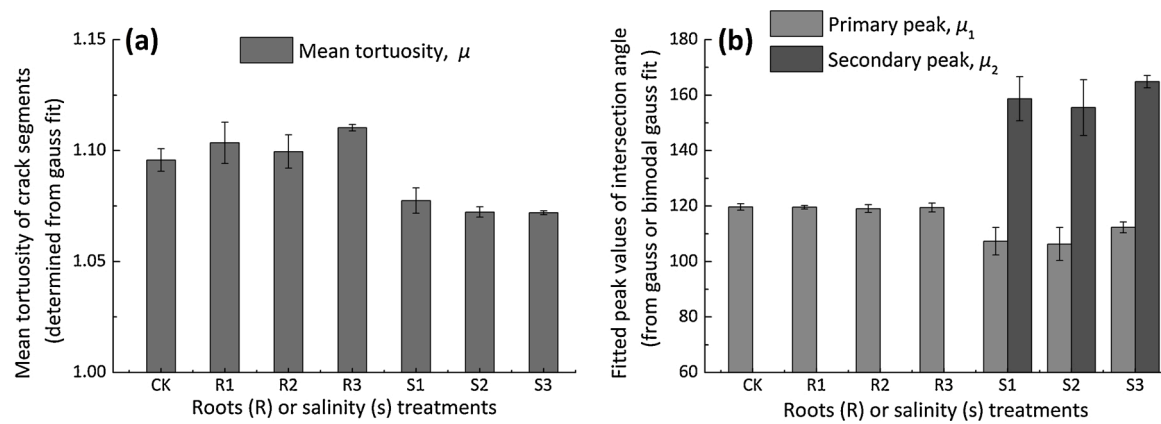


Fig. 8. (a) The average value ($n = 3$ replicates) of means of single-crack tortuosity fitted from Gauss fit; (b) the average value ($n = 3$ replicates) of means of crack intersection angle fitted from Gauss or bimodal Gauss fit. Note that there are two peaks, i.e. primary peak μ_1 and secondary peak μ_2 for S1, S2, and S3 treatments.

Table 4

Summary of Chi-square test for the hypothetical distributions of crack parameters for all root and salinity treatments.

Crack measure	Treat	Sample size	Distribution	Goodness of fit (R^2)	χ^2 value	$\chi^2_{0.05(k-1)}$ value	Accept or reject?
Crack tortuosity	CK-1	204	Normal	0.98	16.12	18.307	Accept
	CK-2	199	Normal	0.93	21.79		(Reject)
	CK-3	272	Normal	0.97	16.70		Accept
	R1-1	288	Normal	0.98	14.50		Accept
	R1-2	252	Normal	0.98	2.43		Accept
	R1-3	226	Normal	0.99	7.33		Accept
	R2-1	364	Normal	0.96	16.25		Accept
	R2-2	214	Normal	0.95	14.29		Accept
	R2-3	218	Normal	0.97	14.19		Accept
	R3-1	252	Normal	0.92	12.81		Accept
	R3-2	321	Normal	0.98	13.29		Accept
	R3-3	256	Normal	0.94	33.79		(Reject)
	S1-1	190	Normal	0.98	17.25		Accept
	S1-2	189	Normal	0.97	16.23		Accept
	S1-3	153	Normal	0.99	1.99		Accept
	S2-1	153	Normal	0.96	13.48		Accept
	S2-2	152	Normal	0.95	25.99		(Reject)
	S2-3	109	Normal	0.95	15.53		Accept
	S3-1	109	Normal	0.99	10.32		Accept
	S3-2	112	Normal	0.98	12.28		Accept
	S3-3	121	Normal	0.97	13.81		Accept
Crack intersection angle	CK-1	237	Normal	0.92	19.98	27.587	Accept
	CK-2	246	Normal	0.95	6.21		Accept
	CK-3	242	Normal	0.95	15.68		Accept
	R1-1	283	Normal	0.93	5.90		Accept
	R1-2	249	Normal	0.93	7.34		Accept
	R1-3	204	Normal	0.94	7.77		Accept
	R2-1	378	Normal	0.90	20.64		Accept
	R2-2	237	Normal	0.96	8.03		Accept
	R2-3	251	Normal	0.99	1.04		Accept
	R3-1	200	Normal	0.94	13.68		Accept
	R3-2	324	Normal	0.96	7.15		Accept
	R3-3	189	Normal	0.93	10.50		Accept
	S1-1	194	Bimodal	0.99	2.33		Accept
	S1-2	237	Bimodal	0.96	4.71		Accept
	S1-3	180	Bimodal	0.97	0.24		Accept
	S2-1	188	Bimodal	0.96	3.02		Accept
	S2-2	159	Bimodal	0.96	4.03		Accept
	S2-3	135	Bimodal	0.95	1.57		Accept
	S3-1	166	Bimodal	0.97	1.17		Accept
	S3-2	156	Bimodal	0.97	1.71		Accept
	S3-3	147	Bimodal	0.98	0.52		Accept

4. Discussion

4.1. Cracking mechanism

The experiments conducted in this study were free desiccation and shrinkage tests without any external mechanical stresses. Soil deformation and shrinkage were governed by the interior soil stresses that

resulted from the decline in the degree of saturation and increasing capillary suction. Macroscopically, soil desiccation cracks were performances of soil deformation and pore system shrinkage (Wang et al., 2018b). For free shrinkage tests, the soil deformation was anisotropic, and exhibited primary-phase vertical subsidence and secondary-phase horizontal-vertical shrinkage, as reported by Peng and Horn (2007), Peron et al. (2009), Shin and Santamarina (2011) and Tollenaar et al.

(2017). Suppose there is an initially thin layer of water ponding on the soil surface, the cracking scenario could be conceptualized as follows: (i) the pore water pressure inside the soil is positive and the capillary pressure is zero, until the water level descends to the soil surface, during which the water constantly evaporates; (ii) the gas-liquid-solid interface descends below the soil surface, and water drains from macropores, resulting in a slight capillary pressure. The shrinkage in this phase shows only vertical subsidence, which corresponds well with the proportional shrinkage phase; (iii) the water drains from smaller and smaller pores and the matric potential becomes increasingly negative. The soil sample shrinks both vertically and horizontally. Desiccating induced negative pore pressures produces compressive effective stress between soil particles because of the developing hydraulic gradient inside the soil profile. Once the volumetric strain overcomes the soil tensile strength in the vicinity where preferential membrane invasion occurs or defects exist, cracks initiate and propagate along the air invasion surface, accompanied by the release of tensile forces. Upon desiccating, the balance between the increase in soil material stiffness and strength, and the generation of increased suctions and tensile stress, gives rise to stain localization and existing crack propagation (Stirling et al., 2017). This phase corresponds to the residual shrinkage phase in which water loss outweighs the decrease in voids volume. Vapor transport inside the soil becomes increasingly difficult; (iv) soil skeleton and crack patterns are nearly stabilized although little evaporation is detected. Little water is trapped in extremely small pores. This matches with the zero shrinkage phase.

4.2. Maize roots effects on cracking patterns

The incorporation of maize roots reduced soil cracks in width, while increased crack length. With roots treated, the soil cracks were more tortuous, and unimodality at 120° was found in the distribution of crack intersection angles. These findings indicate that maize roots can be a potential amendment to restrain soil shrinkage and cracking extent. The root mechanical reinforcement increases 'apparent cohesion' or 'root cohesion', and higher densities of roots result in greater tensile strength of roots system (Bordoloi et al., 2018). Bordoloi et al. (2019) reported that crack intensity factor (CIF) could be significantly reduced by 50 % in natural-fiber-reinforced soils (Bordoloi et al., 2019). Similar conclusions were drawn in undisturbed soils with grass that Bermuda grass in clay soils constrained crack development in comparison with bare soils during wetting-drying cycles (Song et al., 2017). The root amendment mechanism can be described as a physical interaction between soil particles and roots. The root surface was attached by surrounding soil particles, thus enhancing the soil tensile strength. The interfacial shear resistance between soils and roots depends on the configuration and composition of soil particles, effective interface contact area, and root surface roughness (Hejazi et al., 2012). The bonding and friction between roots and soil particles that result from the increasing effective contact area effectively restrict the relative movement between soil matrix and roots, thus restraining soil deformation and shrinkage. This efficacy helps to restrict separation between two soil pedons at a crack aperture. Besides, once cracking was initiated, the roots stretching across the cracks would take on the tension inside the matrix with root-soil friction, due to the roots' stiffness and ductility, which impeded further soil deformation (Jiang et al., 2010; Wang et al., 2018a).

Crack morphology in roots treated soils was quite different from that in CK treatments. The crack apertures in root treated soils were smaller with shorter lengths in contrast to those in CK treatment. The crack network was in a fragmented pattern as affected by root incorporation. Besides, the crack intersection angle changes from '+' to 'Y' shape when the soil was incorporated with roots, according to the observed crack patterns and distribution of crack angle. These findings can be attributed to the altered maximum stress release criterion and crack initiation-propagation criterion induced by root incorporation (Tang et al., 2012). The bridging effects, stiffness, and ductility of roots significantly

affected the stress field in soils and changed the intrinsic release criteria of the tensile stress, resulting in pronounced changes in crack intersection angle and network pattern.

The cracking behavior of root-treated soils also depends on the root additive content, as shown in Fig. 4. This correlates closely with the effective contact area between roots and soil particles. With an increase in roots contents, the effective contact area increases, thus enhancing the friction between the root-soil system and restraining the relative movement between soil matrix and roots (Chaduvula et al., 2017). Although the soils can be effectively reinforced by roots and fibers, there is a threshold for additive content. Beyond this threshold content, the roots or fiber content shows no additional effects on soil strength improvement or cracking patterns, as reported by (Olgun, 2013). Other research has documented that the exudate discharged by living plants roots consists of a broad range of compounds that would exponentially increase the fracture toughness and bonding energy in expansive clayey soils (Zhang et al., 2008). The exudate bonds soil particles and hence aggregates together, thus improving the soil cohesion and resisting cracking mobilization (Bordoloi et al., 2020). This provides a new perspective on how roots restrain cracking in situ fields. These findings suggest that roots additives can reduce crack widths and modify cracking morphology, making it a potential amendment to restrain soil shrinkage and inhibit preferential flow induced by shrinkage cracks, since the crack flow flux is proportional to the cube of the crack width.

4.3. NaCl effects on crack morphology

With increase in NaCl addition, soil crack width, area, and crack network connectivity also increased, whereas crack length tended to decrease. In this study, analytical reagent NaCl was used and the soil contained high amounts of NaCl, signifying an extremely high sodium adsorption ratio (SAR) of the soil. As discussed above, cracking behavior is intrinsically a manifestation of soil deformation or shrinkage, which is governed by the tensile stresses resulting from the increase in capillary suction as evaporation progresses (Zhang et al., 2016). Cracks tend to occur once the suction stress generated in the soil matrix exceeds the bonding strength between soil particles or aggregates where defects exist (Wang et al., 2017). The cracking mechanism in sodic soils is associated with the physical-chemical processes occurring in the soil matrix, depending on the soil clay minerals and salt types. Sodicty changes the cracking mechanism by modifying soil tensile strength, shrinkage characteristics, and hydraulic properties. Sodic soils have a larger diffuse double layer in comparison with the soils dominated by divalent cations, which causes dispersion within soil aggregates or particles. Thus, the water and sodium were prone to enter the inner layers of clays and hydrate soil particles, increasing the spacing and repulsion between clay particles (Klopp, 2015). A high level of exchangeable sodium disrupts the physical composition and aggregation of clay particles (Huang et al., 2019). As a result, the tensile strength of a semi-saturated soil with high SAR or sodicty is reduced, making it more prone to crack propagation (DeCarlo and Shokri, 2014b; Rahimi et al., 2000; Zabot et al., 1997). A sodium hazard that results in a breakdown in the soil's physical structure (Fipps, 2003) would expedite energy release and cracking process.

Crack intersection angle of $90\text{--}100^\circ$ also conforms with this strain energy release theory. As is discussed before, upon drying the strain energy induced by suction stress is released through cracking, and the cracking tip is prone to propagate through the most efficient release path (Kodikara et al., 2000). Cohesive aggregates are decomposed by high sodicty, and this breakdown of clay particles facilitates energy release and cracking process and decreases the crack propagation path, thereby producing less tortuous individual cracks and highly connected networks. Propagating cracks in the cohesive media would seek for the most effective intersecting angle with other cracks under the principle of 'maximum stress release criterion'. Since the internal stress normal to the pre-existing crack wall (free boundary) has already been released,

the propagating crack approaches perpendicularly to the pre-existing crack (Shin and Santamarina, 2011). We conclude that increasing sodicity results in a breakdown of clay particles, aggregates, and decreasing tensile strength in desiccating clay soils, thereby modifying the energy release mechanism, decreasing crack propagation path, and resulting in crack morphology of higher connectivity compared with the CK treatment.

4.4. How convincing can the crack morphology in small-scale samples be generalized to larger samples or in-situ soils?

Although our experimental results provide compelling evidence that addition of roots or salinity markedly modifies cracking intensity and morphology, limitations are worth noting regarding the sample size (e.g. length and layer thickness) and boundary effects (e.g. basal roughness and friction) on the cracking morphology. It is noteworthy that the cracking behaviors in situ are also influenced by decaying roots, worm holes, tillage, gravels or other drivers that leads to soil heterogeneity in farmland. Here we just analyze the effects of size and boundary. The effect of soil layer thickness on crack morphology has been extensively investigated by researchers (Costa et al., 2013; Lakshmikantha et al., 2012; Shorlin et al., 2000; Zeng et al., 2020). It is generally accepted that thin specimens yield greater number of cracks with smaller aggregated clods (cell areas), whereas fewer, wider cracks are typically visible in thicker specimens (Lakshmikantha et al., 2012). Several factors could account for this phenomenon, i.e. size effects (sample length and thickness), shrinkage heterogeneity in top and bottom soil, base-soil interface friction, soil drying rate that results from surface evaporation pull. Also, varying layer thickness gives rise to interplay between interface friction, shrinkage heterogeneity and drying rate.

The behaviors of drying rate and vertical uniformity of drying extent vary with soil layer thickness. These secondary factors differentiate crack morphology, e.g. areas of aggregated clods and crack aperture width. Thinner layer generates a reasonably uniform suction profile, expediting shrinkage and cracking behaviors. Whereas thicker layers make the migration path of pore water longer from the subsoil to the evaporation surface, thus the suction profile does not fully develop into the bottom soil during the cracking process (Zeng et al., 2020). An innovative concept 'effective layer thickness' is proposed, below which the development of suction and tensile stress decelerate. The unaffected subsoil would constrain and delay the cracking process.

Basal friction is another boundary effect that modifies crack morphology. The base interfacial friction is dominated by both interfacial roughness and shrinkage capacity (Zeng et al., 2019). For a thinner soil layer, the evaporation pull rapidly reaches the subsoil and drain the pore water at the base-subsoil interface. Then the soil shrinkage begins and stimulates the basal friction that dominates cracking behaviors. Whereas, for thicker soils, it takes much longer for the drying front to penetrate into the subsoil, resulting in less basal friction that would be triggered by deep soil shrinkage. When the evaporation begins to exert influences on the bottom soils and the tensile stress plays a non-negligible role in the subsoil, the superficial crack skeleton already stabilizes and hardly produces any new cracks. Other findings reported that an increase in interfacial friction (higher roughness) results in denser crack segments, smaller aggregated clods (cell) and longer total crack length (DeCarlo and Shokri, 2014a; Zeng et al., 2019). According to this finding, a bold and plausible assumption was proposed that an absolutely smooth substrate generates no desiccation cracks but wider side-boundary gaps (Zeng et al., 2019).

Cracking behavior in geometrically similar specimens of different size subjected to desiccation is proved to be well described by Bazant's size-effect law (Lakshmikantha et al., 2012), $(\sigma_t/\sigma_{cr})^2 = A(d/d_0) + C$, where σ_{cr} and σ_t are the cracking stress and tensile strength, respectively. d is the specimen dimension (e.g. side length), d_0 is the reference dimension (e.g. maximum particle size), A and C are fitted constant

parameters. Crack length, width and cell areas in cracked soils depend on the superficial area and thickness of the specimens, which can be described by energy related Bazant's size-effect law for the sake of simplicity. The specimen size in our experiments is $22.0 \times 22.0 \times 1.0$ cm², and the lateral and bottom boundaries are Perspex. In the light of the aforementioned findings, the basal friction and size effects play a prominent role in cracking mechanism. Larger aggregated clods could be expected in situ field conditions with the identical soil. The 'effective layer thickness' is an interesting concept, greater than which the soil layer will be faintly influenced by the substrate friction coefficient or the layer thickness. In our experiments, the specimens treated with root densities and salinity contents were all kept in identical containers and atmospheric conditions. What we concern is how root densities and NaCl affect crack intensity and morphology through parallel comparisons. The conclusions that maize roots restrain cracking and NaCl addition induces wider crack apertures with larger aggregated clods could be generalized to in-situ conditions, whereas the crack ratio, width, length and cell areas in-situ and indoor scales under the same treatment are not necessarily the same. There is evidence indicating that the peak values in the distribution of crack intersection angles remain unchanged in varying layer thickness, while the range of the angle distribution may change (Shorlin et al., 2000). Generally, the cracking behaviors in-situ and specimen scales are dimensionally similar. Still, further investigation needs to validate whether crack tortuosity and connectivity of superficial crack network remain unchanged as affected by the size effects of soil pedons.

5. Conclusion

We investigated the physicochemical effects of maize roots and sodium chloride on the cracking dynamics, crack geometry, crack network topology, and statistical features of crack morphology of desiccating soil layers. Results showed that incorporating roots into soils decreased crack area, width, and network connectivity, but increased crack length. A highly fragmented crack pattern was observed with increasing root content. Samples with higher root content had a vast number of dead-end cracks and spikes, and single cracks exhibited higher tortuosity. A unimodality at 120° was found in the distribution of intersection angle for root-treated soils. These findings indicate that maize roots can be a potential amendment to control soil shrinkage and cracking degree, as the bridging effects, stiffness, and ductility of roots effectively increase soil tensile strength and alter the energy release criteria of tensile stress.

The study investigated the effects of salinity (NaCl) on crack morphology in thin soil layers. Results showed that crack morphology was highly dependent on NaCl concentration. With higher salinity, crack area and width increased, whereas length tended to decrease. The salinity treatment resulted in higher connectivity of crack networks and less tortuous crack segment. Distribution of crack intersection angles in sodic soils displayed asymmetric bimodality, with primary and secondary peaks around 105° and 165°, respectively. The observed behaviors associate closely with the breakdown of clay particles or aggregates in soil physical structure, a decline in evaporation, and retardation in soil shrinkage. For all root and salinity treatments, we proposed a statistical function that describes the probability distribution of crack area with respect to crack width classes, at varying water contents.

Our results provide compelling evidence that maize roots and NaCl markedly affect crack morphology and mechanisms, which provides a theoretical basis for means of crack prevention and control of crack preferential flow. However, by parallel comparisons, what we concerned and simply revealed is how root densities and NaCl affect crack intensity and morphology in relatively small-scale samples. The cracking behaviors in-situ soils are highly susceptible to size effects (sample length and layer thickness), substrate condition and tillage methods. We hypothesize that superficial cracking behaviors in specimen and in-situ scales are dimensionally similar, and the generalization

of these conclusions to in-situ scale needs to be further validated.

Declaration of Competing Interest

The authors report no declarations of interest.

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